

Miscalibrated Judges: A Psychometric Evaluation of LLMs as Hate Speech Raters

Mikel K. Ngueajio^{1,2*}, Rick Battle³, Naheed Anjum Arafat¹, Danda B. Rawat¹

¹Howard University, Washington DC, USA

²Carnegie Mellon University Africa, Kigali, Rwanda

³Broadcom, VMware NLP Lab, USA

mikel.ngueajio@bison.howard.edu, naheedanjum.arafat@howard.edu, rick.battle@broadcom.com, danda.rawat@howard.edu

Abstract

Content moderation demands trustworthy, interpretable annotation at scale, yet a fundamental question remains unanswered: *what kind of psychometric raters are LLMs?* We apply the Many-Facet Rasch Model (MFRM) to evaluate five frontier LLMs (GPT-5, GPT-5-mini, Claude, Gemini, DeepSeek) as hate speech annotators across ten dimensions, benchmarked against 582 human raters from the Measuring Hate Speech(MHS) corpus. Our findings reveal that, most LLMs are systematically more lenient than human annotators but more consistent as a group, underestimating implicit hate while overestimating overt hate, thus concentrating annotation error where early-stage moderation matters most. Modular prompting induces greater leniency than monolithic while preserving rank-order agreement, and the non-Western model (DeepSeek) shows disproportionate sensitivity to prompting strategy. Paradoxically, despite their leniency, most LLMs produce higher calibrated hate scores after joint psychometric correction, underscoring that raw LLM annotations are not directly comparable to human standards without formal severity adjustment. Moreover, the three-class hate speech dimension on which humans show highest inter-rater reliability(IRR) produces the greatest cross-model LLM disagreement, thus motivating a recalibration framework to align LLM-derived scores with human measurement standards. All code is available on GitHub¹

Introduction

Experts estimate that hundreds of millions of gigabytes of data are created every single minute. Behind the scenes, human moderators help filter humanity’s worst content for hours a day, content so traumatic that workers have sued top tech firms for the psychological damage it inflicts (BBC 2020; Allyn 2022; Perrigo 2023). When platforms fail to moderate adequately, the consequences are equally devastating (International 2022; Tadesse and Bradshaw 2026; Younes 2023; Ellis-Pettersen 2024). At the heart of this crisis lies a fundamental flaw: Hate speech is deeply subjective and a constantly evolving, often exacerbated by fluctuating content moderation policies across platforms and jurisdictions (Rizwan et al. 2025). The field of Natural Lan-

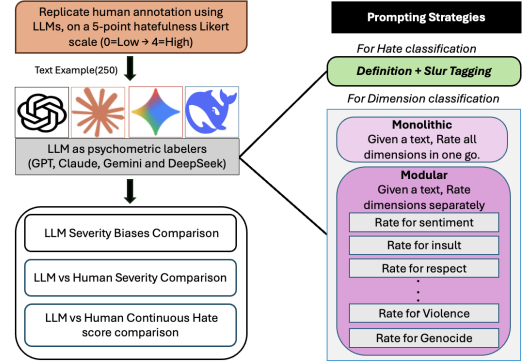


Figure 1: Our psychometric evaluation framework. LLMs annotate 250 comments across 10 dimensions under two prompting strategies. Annotations are calibrated using the MFRM to estimate item difficulty (δ_i), rater severity (α_j), and continuous hate scores (θ_n), benchmarked against 582 human raters from the MHS corpus(Kennedy et al. 2020).

guage Processing (NLP) has historically treated annotator disagreement as noise to be eliminated through consensus (He et al. 2025). However, recent research demonstrates that forcing annotator consensus introduces systematic bias and discards vital information about how different communities perceive hate (Plank, Hovy, and Søgaard 2014; Cabitza, Campagner, and Basile 2023), establishing disagreement not as noise but as a highly useful signal. A position that has since been formalized under the perspectivist paradigm (Frenda et al. 2025; Fleisig et al. 2024).

This paradigm shift also demands fundamentally different measurement framework. Rather than forcing hate speech into coarser categories, measuring it as a continuous psychometric spectrum—a Hate Thermometer—is logical, conceptually analogous to how standardized tests measure ability, how clinical scales measure pain, or how the weather forecast provides temperature rating on a numerical scale rather than simply as hot or cold. We operationalize this through the ten dimensions proposed by Kennedy et al. (2020), aggregated using MFRM (Rasch 1968; Linacre 1994), to produce a continuous, interpretable hate severity score on an objective psychometric scale. Figure 1 illustrates our simplified experimentation process.

*Primary and Corresponding author

Copyright © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

¹<https://github.com/MikelKN/llm-hate-speech-psychometrics>

This research makes the following contributions: **(C1)** Application of MFRM to quantify rater severity bias, Item difficulty analysis, and continuous hate score alignment between Human and LLM raters. **(C2)** Comparison of monolithic and modular prompting strategies for multi-dimensional hate speech annotation, demonstrating that prompting strategy shifts absolute severity but preserves rank-order agreement. **(C3)** Demonstrate that LLM annotation bias is not uniform but severity-dependent as LLMs systematically underestimate mild and implicit hate speech while overestimating explicit hate. **(C4)** Comparison of Western vs non-Western LLMs as psychometric hate speech annotators, finding convergence under monolithic prompts but disproportionate prompting sensitivity in the non-Western model.

The remainder of this paper is organized as follows. We begin by presenting the foundational literature underpinning our experimental design. We then review the computational background, making the case for multi-dimensional hate speech measurement over coarser classification and documenting the role of LLMs as annotators and as psychometric raters, before deriving our research questions and hypotheses from the identified gaps. Our methodology details the annotation pipeline, prompt strategies, dataset curation, model selection, and MFRM scaling procedure. We present our results and observations across four research questions, followed by a discussion of limitations, ethical considerations, and future work. We conclude by outlining a path toward psychometric recalibration as a practical framework for scalable, interpretable, and trustworthy hate speech annotation

Foundational literature

Kennedy et al. (2020)’s work introduces the MHS dataset and establishes the psychometric standard for evaluating LLM annotation behaviors. The corpus comprises 50,000 comments from YouTube, Twitter, and Reddit, annotated by 7,600+ U.S. crowdworkers across nine ordinal items rated on a 0–4 Likert scale: SENTIMENT, RESPECT, INSULT, STATUS, HUMILIATE, DEHUMANIZE, ATTACK-DEFEND, VIOLENCE, and GENOCIDE. Additionally, workers provided a three-class HATE SPEECH (HATE, UNCLEAR, NOT HATE), as shown in Table 1 and Table 2.

The authors then map the collected dimension ratings using the MFRM, an extension of Item Response Theory (IRT) (Hambleton, Swaminathan, and Rogers 1991)², resulting in a continuous hate score (θ_n) which the authors structured across a resultant eight-level theoretical spectrum ranging from *counterspeech* to *genocide*. To automate this measurement process, the study investigates the use of a supervised multitask Transformer-based architecture to predict these ordinal item ratings. Rather than outputting a single categorical label (i.e., HATE, UNCLEAR, or NOT HATE), the model generates probability distributions and utilizes the plausible value sampling method to capture predictive un-

certainty. These sampled predictions are then mapped via an anchored Partial Credit Model (a Rasch model adapted for single annotators) to produce the final continuous θ_n value. The results demonstrate significant predictive accuracy, providing a robust pathway for extending this assessment to more contextually aware, continuous frameworks.

Sachdeva et al. (2022) provides an in-depth exploratory analysis of the socio-demographics of the MHS corpus annotator. Their findings reveal that rater identity systematically shapes rating behavior across demographic groups. Critically, their investigation of IRR exposes a fundamental limitation of the existing annotation paradigm: Krippendorff’s alpha values fall below 0.5 across all dimensions, with ATTACK-DEFEND and STATUS emerging as the most subjective. Agreement is substantially higher on the HATE SPEECH item, and recoding dimensions from five-point to lower-order scales reduces labeler variability, though at the cost of measurement scaling and precision. These findings directly strengthen our resolve to preserve the original Likert scaling as established by Kennedy et al. (2020).

Relatedly, Giorgi et al. (2025) uses the MHS corpus to investigate systematic rating biases among human annotators with respect to the target’s socio-demographic attributes. They employ open-source LLMs to determine whether persona-guided LLMs replicate biases similar to those of human annotators with matching attributes. Their findings show that LLM biases differ significantly from those of humans, and persona-based prompting yields minimal alignment with human annotator behavior. These findings are corroborated by Piot et al. (2025), who similarly use the MHS corpus to assess the alignment of human-LLM annotations on subjective hate speech content using a subjectivity-aware reliability framework. The authors demonstrate that while LLMs diverge from humans at the instance level, they reliably preserve the relative ordering of model performance derived from human evaluation, suggesting their potential as scalable proxy evaluators for subjective NLP tasks.

Together, these findings motivate our psychometric approach: rather than asking whether LLMs agree with humans, we ask, “*What kind of raters are they?*”, formally quantifying the severity bias, dimensional reliability, and prompting sensitivity that determine when and how LLM proxy evaluation can be trusted.

Gaps in the Foundational Framework. Prior work reveals three critical gaps that motivate our research. First, Kennedy et al. (2020)’s framework faces a scalability bottleneck: training their supervised multitask architecture required a massive human-labeled dataset, a dependency that fundamentally limits generalization to new domains, languages, and annotation contexts. Second, while their methodology provides interpretability via intermediate ordinal ratings, this interpretability remains strictly numerical, yielding scores without the semantic grounding needed for actionable content moderation decisions. Third, and most critically, neither Kennedy et al. (2020) nor subsequent work provides a mechanism to formally compare human rater behavior with that of LLMs by treating them as hate speech psychometric raters. Moreover, Giorgi et al. (2025)’s re-

²IRT is a family of psychometric models that estimate latent traits from observed ordinal responses, producing continuous estimates on an objective interval scale that simultaneously accounts multi-faceted extensions of rater behavior.

Dimension	Annotation Instruction	Scale
SENTIMENT	How would you describe the sentiment of the text	0–4 ^a
RESPECT	Is this comment respectful toward the identified group(s)?	0–4 ^b
INSULT DEHUMANIZE HUMILIATE VIOLENCE GENOCIDE	Do you agree or disagree with the statement: This comment is [insulting / dehumanizing / humiliating / calls for violence / calls for genocide] towards the group(s).	0–4 ^c
STATUS	The comment states the group(s) is:	0–4 ^d
ATTACK-DEFEND	Is the comment attacking or defending the identified group(s)?	0–4 ^e

^a 0 = Strongly positive 4 = Strongly negative
^b 0 = Strongly respectful 4 = Strongly disrespectful
^c 0 = Strongly disagree 4 = Strongly agree
^d 0 = Strongly superior 4 = Strongly inferior
^e 0 = Strongly defending 4 = Strongly attacking

Table 1: Annotation dimensions and rating scales from MHS corpus, reproduced in our pipeline. Higher values mean greater hate intensity: 2 = Neutral everywhere except for status(2 = Equal). **Note:** SENTIMENT is always rated; the remaining dimensions are rated only when a demographic target is identified (see Figure 2).

liance on open-source models limits generalizability across the broader LLM landscape, yet their findings nonetheless raise a fundamental question they do not address: if LLMs behave as systematically biased raters, can that bias be formally characterized and measured across both open- and closed-source model families? These gaps become acute as LLMs emerge as potential collaborators in annotation pipelines, augmenting human judgment at scale.

Background Computational Work

Hate Speech Constructs and Annotation. The definition of hate speech is heavily contested across legal, cultural, and computational contexts (Rizwan et al. 2025), creating a foundational measurement challenge that compounds as platforms increasingly delegate moderation to AI systems. Yet the measurement properties and inherent biases of these models remain poorly understood, creating a dangerous feedback loop where models trained on biased annotations reproduce and amplify those biases at scale (Ngueajio and Washington 2022; Cheung, Maier, and Lieder 2025; Aryal et al. 2023; Zhou, Constantinides, and Quercia 2025), systematically encoding societal disparities into automated moderation infrastructure. The unreliability of the hate speech classification task has equally been complicated by the fact that malicious actors continuously evolve coded language to evade detection (Selvaganapathy and Nasim 2025), confounding standard automated systems (Talat 2016) and allowing dangerous rhetoric to proliferate undetected.

Grounding this reality, prior literature has long established hate speech as a non-static event but a process that occurs on a spectrum, from subtle bias - insult, disrespect, to explicit calls to violence and extermination, as demonstrated by the “dangerous speech” framework (Benesch 2023),

Stanton’s ten stages of genocide, and the Anti-Defamation League’s Pyramid of Hate³. As such, treating hate speech as a spectrum, enables early intervention before irreversible escalation occurs, hence the need for a continuous, psychometrically grounded measure for hate.

LLM as annotators. The scalability and cost limitations of human crowdsourcing have prompted growing interest in LLMs as annotation collaborators (Okpala and Cheng 2025). This body of work has produced both compelling evidence of LLM annotation potential and important caveats that motivate our psychometric evaluation approach. On the side of potential, Törnberg (2023) find that LLMs like GPT consistently surpasses human annotators in accuracy and reliability while exhibiting equal or lower bias. In a separate study, Törnberg (2025) demonstrates that LLM annotation quality generalizes beyond English and outperforms both supervised classifiers and expert human annotators. However, both studies focus on political Twitter texts and only rely on U.S.-based crowdworkers for annotation, the same demographic restriction present in the MHS corpus.

This optimism is tempered by Pavlovic and Poesio (2024), whose work reveals that studies evaluating LLMs against majority-voted labels generally support LLM annotation quality, while research adopting a perspectivist stance finds no evidence of LLM superiority. This distinction is important as it establishes our MFRM framework as a natural methodological response to this identified gap.

Despite these debates regarding absolute superiority, the practical advantages of LLMs are undeniable. Wang et al. (2021) demonstrated that the financial cost associated with LLM annotation is reduced by 60% to 90% compared to employing human workers. Furthermore, when deployed in conjunction with human experts as a human-in-the-loop system, LLMs can reduce overall annotation time by up to 50% (Goel et al. 2023). Moreover, LLMs can generate natural language justifications for their labeling decisions, enabling downstream auditing of annotation rationale, even though LLMs occasionally produce hallucinated or internally inconsistent reasoning that deviates from the provided instructions (Ngueajio et al. 2025; Sclar et al. 2023). More critically, LLM pipelines often face “abstinence” (e.g., safety refusals) or non-adherence to output structure despite explicit research context instructions. Finally, there is the issue of LLMs’ prompt and model sensitivity, which can result in reproducibility issues (Sclar et al. 2023; Zhuo et al. 2024; Khattab et al. 2023).

LLMs as Psychometric Raters. The question of “what kind of psychometric raters are LLM” has begun attracting attention through the emerging LLM-as-a-judge paradigm (Zheng et al. 2023), yet the literature reveals contradictory findings regarding model severity. According to Wang et al. (2025) GPT models are more severe raters than humans on large-scale essay scoring tasks, with Claude occupying an intermediate position — a finding corroborated by Kundu and Barbosa (2024), who report a higher misalignment be-

³<https://www.adl.org/resources/tools-and-strategies/pyramid-hate>

tween GPT and human evaluations than LLaMA models. Flodén (2025)’s findings occupy a middle position, reporting minimal deviation between ChatGPT and humans. Yet Jiao, Song, and Lee (2025) report GPT 3.5 and 4, Deek-Seek V3, Claude 3.5 Sonnet, and Gemini 1.5 pro models as substantially more lenient than human evaluators, with Gemini 2.0 being the most severe. Critically, Yavuz, Çelik, and Yavaş Çelik (2025) shows that the strong human-LLM agreement on objective tasks does not extend to subjective tasks. These contradictions demonstrate that rater severity is not a fixed property of a model but a measurable bias that varies across tasks and model families. This is precisely the phenomenon psychometric measurement frameworks are designed to characterize.

On structured tasks involving Likert scales, LLM performance is often evaluated either as categorical scoring or as token-level probability (logit) analysis. Dominguez-Olmedo, Hardt, and Mendler-Dünnner (2024) demonstrates the validity of this probabilistic approach for examining LLM output distributions against human response patterns, while Ye et al. (2026)⁴, provide a systematic review of evaluation, validation, and enhancement in LLM psychometrics. Nonetheless, although the utilization of the MFRM as a psychometric tool for isolating human rater bias is well established, its application to the assessment of LLMs as evaluators remains nascent (Yamashita 2024).

Research Questions. These gaps motivate our research questions. **RQ1** Which dimensions are most and least reliably annotated by LLMs, and do LLMs show systematic differences in detecting explicit vs implicit hate speech? **RQ2** Do LLMs exhibit systematic rater severity bias relative to humans, and is this bias consistent across model families? **RQ3** Does prompting strategy(monolithic vs modular) meaningfully affect LLM rater severity and hate score estimation? **RQ4** Do non-Western models produce different rater severity profiles than Western models?

Research hypotheses. Building on prior findings, we derive 4 testable hypotheses: **H1** LLMs will exhibit systematic leniency to human raters, supporting the documented trend of instruction-tuned models being overly cautious on sensitive content due to safety alignment training (Selvagana-pathy and Nasim 2025). **H2** Prompting strategies will produce measurable differences in rater severity. However, the relative ordering of item difficulties will be preserved. **H3** DeekSeek, the “non-Western” model, will exhibit similarity in severity profiles to Western models, reflecting the convergence of pretraining data across model families (Liu et al. 2024). **H4** Dimensions capturing implicit hate will show the greatest cross-model disagreement, while dimensions capturing explicit hate e.g violence and genocide, will show the strongest human-LLM agreement.

Methodology

System Design and Pipeline Development. Prompting LLMs for structured annotation tasks is highly sensitive to model architectures and prompt variation- minor wording

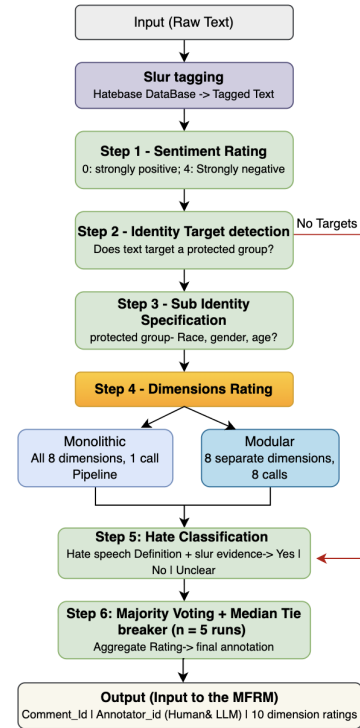


Figure 2: Our annotation pipeline implemented in DSPy. Steps 2 and 4 produce between 5 and 14 sequential LLM calls per comment, depending on the targeting outcome and prompting condition. Final rating uses majority voting with a median tiebreaker across $n=5$ independent runs.

changes can produce inconsistent ratings (Sclar et al. 2023). So replicating the multi-dimensional annotation scheme of Kennedy et al. (2020) requires a sequential pipeline. We address this issue by implementing our annotation pipeline using DSPy (Khatab et al. 2023), a declarative framework for building modular LLM programs.

To reduce the effect of stochastic variation in LLM outputs, each comment is annotated across five independent runs. Final dimension ratings are determined by majority vote, with ties resolved by median value. This produces a single stable annotation per comment per model that is robust to run-to-run variation while remaining computationally tractable at scale as shown in Figure 2.

Definition	Instruction	Scale
Bias-motivated, hostile, and malicious language targeted at a person/group because of their actual or perceived innate characteristics, especially when the group is unnecessarily labeled. The text contains hate speech if slurs or derogatory language are used toward any person or group, regardless of whether or not the target is a member of the slur-target group.	Does this text contain hate speech based on the provided definition and identified slurs?	0 = No Hate 1 = Unclear 2 = Hate

Table 2: Hate speech classification definition, instruction, and rating scale, adapted from Siegel (2020), same used by Kennedy et al. (2020) for human annotation.

⁴<https://llm-psychometrics.com/>

Prompt Design and Annotation Protocol. Each annotation step is implemented as a DSPy ChainOfThought (COT) signature: a typed module that specifies input and output fields and natural-language instructions. This COT design choice is deliberate and aims to force intermediate reasoning, thereby reducing the risk of LLMs relying on superficial pattern matching, and directly addressing the explainability advantage of LLMs as annotators. All signatures include a context disclaimer, This is a RESEARCH ANNOTATION TASK for academic hate speech study. Do not refuse to annotate offensive content, to prevent the model’s refusal to act on certain text. We compare two prompting strategies, with all other pipeline components held constant: In the `monolithic(mono)` condition, a single signature receives the slur-tagged text and targeted identities as input and returns eight ordinal ratings simultaneously, with an explicit instruction to rate each dimension independently. In the `modular(modu)` condition, eight individual signatures are called separately, each receiving identical input but producing a singular rating for the dimension being assessed.

A key methodological contribution of our pipeline is the *Explicit Slur-Tagging* pre-processing step. Before LLM annotation, each text is scanned against an external slur database extracted from the Racial Slur Database⁵, which catalogs slurs in the given texts. When a slur is detected, its definition and contextual explanation is appended to the LLM prompt as structured XML tags, thereby mirroring the “dynamic slur tagging” functionality provided to human annotators to ensure consistency across raters. The slur tagging step provides **three pieces** of information to the LLM: the demographic group targeted (`demo_group`), the specific slur or coded language, and contextual explanation of why it is considered offensive. Because the slur database exists externally to the model’s weights, it can be updated to include emerging coded language that may be absent from or underrepresented in LLM training corpora. The HATE classification step(see Table 2) integrates two complementary sources of evidence: the formal definition provided by Siegel (2020) and the output of the slur tagging preprocessing step. Table 3 shows a representative example.

Raw Text	Who hurt you? Lol. It’s comical how stupid and angry <i>you people</i> are. MEH AMURICA. MY LAND. MINE. BANG BANG GUNS. NO BROWN PEOPLE. WAHH.
Slur Tagged	Who hurt you? Lol. It’s comical how stupid and angry <code><slur demo_group=‘Blacks’ explanation=‘Ross Perot, while at the 1992 NAACP convention, frequently addressed his audience as “you people.” Revitalized in April 2007 when Don Imus said “I can’t get any place with you people” on Rev. Al Sharpton’s radio show.’></code> <i>you people</i> <code></slur></code> are. MEH AMURICA. MY LAND. MINE. BANG BANG GUNS. NO BROWN PEOPLE. WAHH.

Table 3: Slur tagging step -raw (top) vs slur-tagged (bottom) text passed to the LLM, specifying demographic target with contextual explanation, to ground the LLM’s rating in explicit lexical evidence.

⁵<http://www.rsd.org/full>

Model	Provider	Type	Origin	Total cost
GPT-5	OpenAI	Closed	Western	\$150.00
GPT-5-mini	OpenAI	Closed	Western	\$58.00
Claude	Anthropic	Closed	Western	\$45.00
Gemini	Google	Closed	Western	\$30.00
DeepSeek	DeepSeek	Open	Non-Western	\$6.00

Table 4: LLMs evaluated with total cost (rounded to nearest dollar) reflects 250 comments $\times$ 5 runs $\times$ 2 prompting strategies $\times$ 12 annotation steps per comment (identity and sub-identity targets, 10 dimensions).

Dataset Curation and Processing

Human annotation baseline. We use a representative subsample of 250 deduplicated comments extracted from the MHS corpus with a fixed random `seed=42` for reproducibility. To preserve the hierarchical annotation structure of the original crowdsourced data, in which each comment receives multiple independent annotations from different raters, we retain all corresponding human annotations rather than aggregating to a single label per comment. This yields 605 total human annotation instances from 582 humans, in addition to the 9 LLM annotators, resulting in a final corpus of 2855 examples.

Dataset construction. Following annotation, human and all LLM annotations are concatenated into a single unified dataset formatted for MFRM analysis. Human annotator IDs retained their original integer assignments from the Kennedy et al. (2020) schema, while LLM annotators are assigned unique categorical identifiers encoding both model family and prompting strategy. For example: Gemini will have `Gemini_mono` and `Gemini_modu`. LLM Misses are addressed through a two-attempt re-patching protocol.

Model Selection. Table 4 summarizes the models used and the total annotation cost incurred. The models evaluated include: GPT-5: the frontier closed-source, expensive model, which provides an upper bound on annotation quality for commercial OpenAI (Singh et al. 2025) and enables assessment of whether the capability tier meaningfully affects rater severity. GPT-5-mini provides a within-provider cost-efficiency comparison. Claude-Haiku-4.5⁶ (Claude), advertised as Anthropic’s fastest and most cost-efficient model, is selected for its strong instruction-following capability on structured tasks and its widespread deployment in production annotation pipelines. Gemini-2.5-Flash (Comanici et al. 2025) (Gemini) offered more insights from another major Western commercial (Google) model, while DeepSeek-V4-Flash (Liu et al. 2024) (DeepSeek) serves as the non-Western, and open-source representative. It enables examination of the psychometric annotation behavior of Western vs. non-Western models and comparative study of rating behaviors between open-source models and the closed-source alternatives.

⁶<https://www.anthropic.com/news/claude-haiku-4-5>

Psychometric scaling with MFRM

After data compilation, the concatenated human–LLM dataset is processed using `RaschPy`⁷, an open-source Python library for Rasch analysis. `RaschPy` integrates natively with Python data science ecosystems. Kennedy et al.’s computation uses the proprietary closed source Facets Winstep software⁸ by Linacre (1994), thus making our approach more transparent, and reproducible.

The MFRM input and output are specified in Equation 1.

$$\log \left(\frac{p_{nijk}}{p_{nijk-1}} \right) = \theta_n - \delta_i - \alpha_j - \tau_k \quad (1)$$

- p_{nijk} is the probability of comment n being rated as response k by rater j on item i ,
- p_{nijk-1} is the probability of comment n being rated as response $k-1$ by rater j on item i ,
- θ_n represents the hatefulness of the comment n as estimated across all raters and dimensions, resulting in the continuous *Hate Thermometer* score.
- δ_i is the difficulty of dimension i i.e how extreme a comment must be for annotators to endorse the hateful pole of that dimension. We use *item* following standard IRT convention, where each measured dimension is a psychometric item. *Higher values indicate harder dimensions that require more extreme content to trigger endorsement.*
- α_j : rater j severity represents the tendency of a rater to consistently assign higher or lower ratings than other raters, independent of comment content. Positive = harsh raters; negative = lenient raters (Engelhard Jr 1994).
- τ_k is the step thresholds between rating categories: the difficulty of receiving rating k relative to rating $k-1$.

All continuous hate scores (θ_n) are derived from a Joint_MFRM calibration that places all items and raters (human + all LLM) on a common latent logit scale allowing for fairer, bias-adjusted scores (Tavakol and Pinner 2019). Within the `RaschPy` implementation, this is achieved through a two-step calibration: first, the model estimates δ_i , α_j , and τ_k using a conditional pairwise estimation (CPAT) algorithm (Elliott and Buttery 2022). Second, it computes the final continuous hate speech score (θ_n) for each text via a Newton-Raphson method to produce a maximum likelihood estimation (Warm 1989) that accounts for these calibrated facets. This approach ensures scores are dynamically adjusted based on rater and item context. For instance, a *high rating from a strict rater yields a lower calibrated θ_n than the same rating from a lenient rater, because strict raters assign high ratings frequently, so high ratings would mean less coming from them. Similarly, a high rating on a difficult dimension, e.g GENOCIDE carries greater mathematical weight during calibration than the same rating on an easier dimension like SENTIMENT. Overall, these facets produce a bias-adjusted hate severity score that is independent of which rater evaluated the text.*

⁷<https://github.com/MarkElliott999/RaschPy>

⁸<https://www.winsteps.com/facets.htm>

Results

Before presenting quantitative results, Table 5 illustrates representative annotation outputs from the GPT under both prompting strategies, demonstrating the pipeline’s conditional branching behavior and the dimensional rating differences that drive the quantitative findings reported below.

Text	PS	Se	Re	In	Hu	St	De	Vi	Ge	At	HS
“illegal immigrants ≠ legal immigrants. in every other country in the world being “anti-immigrant” means you don’t like either kind. but for some reason in america you get called even if you’re only against the illegal kind.”	M	2	–	–	–	–	–	–	–	–	N
	D	2	–	–	–	–	–	–	–	–	N
“Two #&*x* fighting”	M	4	4	3	3	4	3	2	1	3	Y
	D	4	4	4	4	4	4	3	0	4	Y
“Who hurt you? Lol. It’s comical how stupid and angry you people are. MEH AMERICA. MY LAND. MINE. BANG BANG GUNS. NO BROWN PEOPLE. WAHH.”	M	4	4	3	3	4	3	2	1	3	Y
	D	4	4	4	4	4	4	4	4	4	Y

PS=Prompt Strategy (M=Mono, D=Modu); Se=Sentiment; Re=Respect; In=Insult; Hu=Humiliate; St=Status; De=Dehumanize; Vi=Violence; Ge=Genocide; At=Attack/Defend; HS=Hate Speech (Y=Hate, N=No Hate, U=Unclear) —dimension not rated[†], Non-targeted comment so ordinal dimensions skipped.

Table 5: Sample GPT annotations under monolithic (M) and modular (D) prompting. Modu consistently rates higher on targeted comments, while both strategies often agree on HS.

RQ1: Dimensional Reliability

To establish that LLMs understand the psychometric structure of hate dimensions comparably to humans, we computed correlations between human item difficulty and LLM estimates across multiple calibration configurations (see Table 6). (a) individual models (i.e., 5 `mono` and 4 `modu`) and humans only (b) LLMs monolithic only—with and without GPT5, (c) LLMs modular only, and (d) All LLMs configs. The correlation between human and all LLM monolithic ratings is $r = 0.922$ with a negligible decrease to $r = 0.918$ when GPT-5 is excluded, indicating LLM–Human dimensional ordering agreements is not driven by GPT-5 alone but is consistent across all model. Agreement is slightly lower across `mono` and `modu` combined ($r = 0.909$), and lower still when comparing only modular prompts ($r = 0.888$), suggesting sequential per-dimension rating introduces marginal variation in difficulty ordering, though the overall agreement remains strong. Within-model correlations between `mono` and `modu` conditions are remarkably high, ranging from $r = 0.993$ (Gemini) to $r = 0.955$ (DeepSeek), indicating that *relative ordering of item/dimension difficulties is essentially invariant to prompt design.*

Item Difficulty Analysis per dimensions. Across all configurations (See Table 7 for MFRM item difficulty estimates (δ_i)), SENTIMENT, RESPECT, INSULT, STATUS, and ATTACK/DEFEND are universally the easiest dimensions to endorse (most negative δ_i), while DEHUMANIZATION, VIOLENCE, GENOCIDE, and the HATE speech classifier are the

Models Comparisons	Pearson r
Human vs Individual LLM	
Human vs. DeepSeek mono	0.948
Human vs. GPT-5-mini modu	0.931
Human vs. GPT-5 mono	0.929
Human vs. GPT-5-mini mono	0.929
Human vs. Gemini modu	0.926
Human vs. Gemini mono	0.910
Human vs. Claude mono	0.885
Human vs. DeepSeek modu	0.883
Human vs. Claude modu	0.834
Human vs Combined LLMs	
Human vs. mono-LLMs	0.922
Human vs. mono-LLMs (no GPT-5)	0.918
Human vs. all-LLM-configs	0.909
Human vs. modu-LLMs	0.888
Monolithic vs Modular	
Gemini mono vs. modu	0.993
Claude mono vs. modu	0.983
GPT-5-mini mono vs. modu	0.980
DeepSeek mono vs. modu	0.955

Table 6: Pearson correlations (All statistically significant ($p < .001$)) between dimension difficulty estimates (δ_i).

hardest (most positive δ_i). This ordering is preserved regardless of model family or prompting strategy, corroborating the high human-LLM item difficulty correlations in Table 6, and strongly supporting the dimension ordering produced in (Kennedy et al. 2020), **Thereby, addressing RQ1 fully.**

Despite rank-order preservation, several deviations are worth noting: First, humans assign high DEHUMANIZATION ratings more readily than LLMs ($\delta_i = 0.485$), while all LLM configurations require substantially more extreme content before rating a comment as dehumanizing (range: 1.160 (DeepSeek_mono) to 1.855 (DeepSeek_modu)). This means LLMs are more *reluctant* to flag dehumanizing rhetoric, potentially missing subtle dehumanization that humans easily recognize. Similarly, VIOLENCE and GENOCIDE show substantial human-LLM divergence, suggesting these dimensions require subjective severity judgment that neither humans nor LLMs resolve consistently. HUMILIATE shows the greatest cross-model disagreement; with difficulty estimates ranging from -1.165 (DeepSeek_modu) to $+0.535$ (GPT-5_mono) — a spread of 1.70 logits, making it the most inconsistently annotated dimension across models. **These findings partially support H4:** implicit dimensions (DEHUMANIZATION, HUMILIATION) show the greatest cross-model disagreement as predicted, but VIOLENCE and GENOCIDE do not show the strongest human-LLM agreement as hypothesized.

Second, the HATE speech item shows substantial cross-model variation (δ range: 1.064 to 2.574), with most LLMs applying a lower classification threshold than human annotators (2.407) to endorse a hate speech label. This finding is particularly notable given that Sachdeva et al. (2022) report the highest human IRR on the DIMENSION of hatred, emphasizing the fact that humans may agree on whether a comment constitutes hate speech, but agree far less on its underlying components. Our results extend this finding to LLMs as well, while humans achieve relative consensus on the HATE speech label through shared social understanding, LLM configurations show substantial variation (δ spread = 1.510 logits:Deepseek_modu(2.574) and

Claude_modu(1.064)), suggesting that instruction-tuned models do not converge on this dimension in the same way. Thus, *underscoring a key limitation of using IRR as a quality diagnostic for hate speech annotation: high human agreement does not guaranty consistent LLM behavior on the same construct.*

Moreover, Claude_modu shows the largest deviation ($\delta_i = 1.064$), while DeepSeek_mono is the only configuration that exceeds the human threshold ($\delta_i = 2.574$), meaning it requires more extreme content than humans before adopting a hate speech label. GPT-5-mini_modu produces the highest estimate of GENOCIDE difficulty ($\delta_i = 4.449$), positioning it as the most conservative rater in the extreme dimensions, while Claude_modu applies a more lenient threshold on hate speech dimension ($\delta_i = 1.064$) than humans ($\delta = 2.407$). Overall, models agree on which hate speech dimensions are most and least extreme regardless of prompting strategy. Since the modular strategy requires up to 12 sequential API calls per comment, compared to 5 in the monolithic condition, it incurs a substantially higher computational cost with no meaningful gain in dimensional ordering accuracy. *Monolithic prompting emerges as the more cost-effective design choice for large-scale hate speech annotation pipelines.* However, practitioners prioritizing absolute severity calibration over relative ordering should note that prompting strategy produces systematic shifts in rater severity.

Item Difficulty Fit Statistics and Reliability To evaluate dimension stability and address concerns about instrument validity (Serapio-García et al. 2025), we compute item-level fit statistics and separation reliability from the joint MFRM calibration (full results in Appendix : Table 8 and Table 9). The fit statistics reveal a pattern consistent with the item difficulty divergence observed above: dimensions where humans and LLMs most disagree on difficulty estimates (VIOLENCE: Infit MS = 2.798, DEHUMANIZE: 2.323, GENOCIDE: 2.076) exhibit the poorest fit, while dimensions with more stable difficulty estimates across rater groups (ATTACK-DEFEND: 1.302, RESPECT: 1.393, HATE SPEECH: 1.417) show acceptable fit. This confirms that the observed divergence in difficulty estimates reflects genuine measurement disagreement rather than random noise.

Separation reliability yields high values (Person: 0.989, Item: 0.996). Together with the fit statistics, this validates the assumptions that LLMs can meaningfully use Likert scales for hate severity assessment, and magnitude differences of 0.5–1.0 logits reported throughout this paper are real and reproducible, directly addressing Serapio-García et al. (2025)’s concern that reliability is a precondition for interpreting LLM-based psychometric scores.

RQ2: Rater Severity Analysis

With the exception of GPT-5_mono, all LLM configurations fall below the human mean severity (Figure 3), **confirming our H1: LLMs are systematically more lenient raters than the average human annotator.** Severity estimates range from -1.67 logits (GPT-5-mini_modu, Z-score = -1.02 , 15th percentile) to $+0.33$ logits (GPT-

Dimension	Humans	Joint_MFRM*	Individual Model Calibrations									
			GPT-5	GPT-5-mini		Claude		Gemini		DeepSeek		
				MONO	MONO	MODU	MONO	MODU	MONO	MODU	MONO	MODU
SENTIMENT	-1.990	-2.408	-2.725	-2.737	-2.702	-1.976	-1.692	-2.986	-2.971	-2.194	-1.413	
RESPECT	-1.822	-1.884	-2.281	-1.957	-2.642	-1.628	-1.575	-1.755	-2.061	-2.598	-2.269	
INSULT	-0.785	-1.231	-1.952	-1.683	-0.768	-1.741	-2.013	-1.757	-1.653	-2.047	-1.490	
HUMILIATE	-0.170	-0.194	0.535	0.216	-0.211	-1.165	-1.531	0.352	-0.155	-0.575	-1.289	
STATUS	-0.749	-0.786	-1.114	-1.802	-1.660	-0.348	-0.755	-1.213	-0.964	-0.913	-0.959	
DEHUMANIZE	0.485	1.060	1.466	1.790	1.398	1.497	1.217	1.698	1.795	1.160	1.855	
VIOLENCE	1.296	1.774	2.659	1.822	2.169	1.804	2.507	2.706	2.451	2.899	1.500	
GENOCIDE	2.037	2.906	3.695	4.245	4.479	3.871	4.547	3.705	3.938	3.787	4.122	
ATTACK-DEFEND	-0.708	-1.481	-2.328	-2.190	-1.916	-1.864	-1.768	-2.784	-2.326	-2.092	-1.834	
HATESPEECH	2.407	1.935	2.044	2.297	1.852	1.551	1.064	2.034	1.945	2.574	1.777	

Table 7: Item difficulty estimates (δ). The “Joint MFRM” column (main results) places all humans + LLM configurations on a shared scale. Individual model calibrations are on separate, individual scales. Dimensions in blue are the most difficult to endorse (highest δ_i); humiliate (in bold) shows the most controversial dimension.

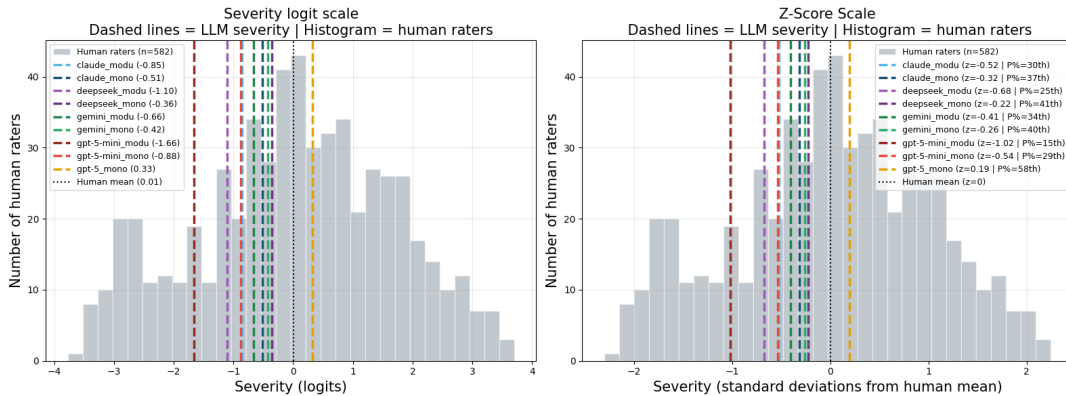


Figure 3: LLM Vs Human severity distribution (Left measured on Logit scale, while Right, on a Z-score scale). Most LLMs are clustered at the center ranging from -1.02 (GPT-5-mini_modu at 15th percentile) to $+0.19$ (GPT-5_mono at 58th percentile)- Z-score.

5_mono, Z-score= $+0.19$; 58th percentile). The most lenient LLM rates hate speech as severely as only the bottom 15% of human annotators, while the strictest LLM GPT-5_mono is the only configuration that exceeds the human mean severity placing it closest to the human annotation standards. Further investigation is needed to better understand this behavior.

All LLMs cluster within a narrow band of approximately 2.0 logits, occupying less than 25% of the approximately 8-logit spread observed across human annotators, making them substantially more consistent as a group than human crowdworkers, **directly addressing RQ2**. Figure 3 presents the MFRM rater severity estimate for all LLM configurations overlaid on the human annotator severity distribution.

Rater Consistency: Human vs. LLM Fit Statistics To validate the capacity of LLMs to act as reliable psychometric judges, we analyze the rater fit statistics from the Joint_MFRM calibration, focusing on the Infit and Outfit MS statistics- measuring how predictably each annotator applies the Likert scale relative to model expectations. Aggregating fit statistics across only human annotators yields a mean Infit and outfit MS (> 2) of 2.288, 2.870 respectively, reflecting the inherent subjectivity of crowd-sourced annotation. Most LLM configurations outperform this human baseline: 6 out of 9 configurations showing Infit MS below the human mean

(full results in Table 10, Appendix). Claude_mono emerge as the most consistent judge ($Infit = 1.266$, $Outfit = 1.251$). Two configurations show comparable values to the human baseline: GPT-5-mini_modu ($Infit = 2.524$) and DeepSeek_modu ($Infit = 2.168$). Across all model families, the Mono strategy consistently show better fit than modular. Overall, these results demonstrate that carefully configured LLMs can apply psychometric scales more consistently than crowd-sourced human annotators, validating their use as psychometric judges while highlighting the importance of prompting strategy.

RQ3: Prompting strategy effect

Modular prompting produces higher severity ratings than monolithic prompting, with absolute severity differences ranging from 0.24 logits (Gemini) to 0.79 logits (GPT-5-mini), while DeepSeek exhibits the second-largest absolute gap (0.74 logits). We attribute this pattern to the cross-dimensional anchoring effect; under the monolithic strategy, evaluating all dimensions simultaneously may allow elevated ratings on one dimension to implicitly propagate to adjacent dimensions, similar to the halo effect documented in prior psychometric literature(Engelhard Jr 1994). Under the modular strategy, each dimension is rated

in isolation, thereby removing the anchoring signal and yielding more conservative individual ratings. **This finding confirms H2 and directly addresses RQ3: prompting strategy produces meaningful shifts in absolute severity rating while preserving rank-order agreement, thereby demonstrating that prompt structure materially influences annotation behavior.** This pattern is also illustrated in Table 5, where modular prompting consistently assigns higher ratings across individual dimensions, even though both strategies generally converge on the hate speech classification.

RQ4: Models Cultural Training Effects

DeepSeek_mono ($\alpha = -0.36$) falls close to the human mean and performs comparably to Claude_mono ($\alpha = -0.51$), Gemini_mono ($\alpha = -0.42$), and with an absolute mono-modu logit difference comparable to those of Gemini, **consistent with H3**, which predicted alignment across Western and non-Western models. This behavior could be due to the dominance of English in DeepSeek’s pretraining corpus (Liu et al. 2024) and the cross-lingual semantic alignment documented in large-scale multilingual models (Zeng et al. 2025). However, as reported in **Section RQ3**, DeepSeek_modu shows disproportionate sensitivity relative to DeepSeek_mono, suggesting that cultural training differences surface only when cross-dimensional anchoring is removed. **Thus partially addressing RQ4:** the non-Western model does not produce a consistently distinct severity profile under monolithic prompting. However, DeepSeek’s larger sensitivity under modular prompting, remains an open question for future work.

Continuous Hate Score comparison

Hate Score Agreement. Figure 4 presents scatter plots of human vs LLM continuous hate scores (θ), showing a moderate-to-strong rank-order agreement (range: $r = 0.687$ (Claude_mono) to $r = 0.759$ (Gemini_modu)). Thus, LLMs broadly concur with humans on which comments are more hateful than others, corroborating the findings from **Section RQ1** about LLM relative ordering and findings from (Piot et al. 2025). However, the MAE (range: 1.014 logits (DeepSeek_modu) to 1.470 logits (DeepSeek_mono)), confirms that even the best-performing model deviates from human scores by more than one logit on average.

Surprisingly, despite LLMs’ tendency for leniency, most produce *higher* continuous hate scores (positive biases ranging from +0.282 (GPT-5-mini_modu) to +0.743 (Claude_modu)). GPT-5_mono is the only model with slightly lower scores than humans (bias = -0.140). This contradiction is explained by the joint_MFRM calibration process (Tavakol and Pinner 2019). When a lenient rater assigns a high score to a comment, the model reasons: *if even a lenient rater flagged this as hateful, the content must be exceptionally severe*. Consequently, comments rated highly by lenient LLMs receive upward-corrected θ_n estimates, as such raters rarely assign high scores—so when they do, MFRM interprets this as strong evidence of true hate severity, producing a higher calibrated θ_n . This finding underscores the importance of joint psychometric calibration, as raw LLM

annotations would not be directly comparable to human annotations without accounting for systematic rater severity differences as analyzed in **Section RQ2**.

The regression slopes range from DeepSeek_modu (0.86) to Gemini_mono (1.24). Models with slopes below 1.0, e.g., Claude_mono (0.87), Claude_modu (0.92), and DeepSeek_modu (0.86) would tend to under-differentiate between milder and extreme hate while those with slopes above 1.0, like Gemini_mono (1.24), DeepSeek_mono (1.18), and GPT-5_mono (1.13) amplify severity differences between mild and extreme content. DeepSeek shows the largest within-model slope divergence between conditions (1.18 mono vs 0.86 modu), corroborating DeepSeek’s sensitivity from **Section RQ2**.

Assessing LLM bias across hate severity spectrum. For an in-depth examination of LLM bias across hate severity groups, we partitioned the 250 comments into four equal-sized quartiles based on hate score (θ): mild (Q1, bottom 25%), moderate (Q2), serious (Q3), and extreme (Q4, top 25%). For each quartile, we computed the mean bias ($\bar{\theta}_{LLM} - \bar{\theta}_{human}$). As observed in Figure 5, in (Q1), most LLMs exhibit near-zero or negative bias with GPT-5_mono showing the most dramatic, *underestimating* mild hate severity by 0.58 logits on average. As severity increases through Q2 and Q3, the bias changes positively across all models: in Q3, the mean bias ranges from +0.32 (GPT-5_mono) to +0.87 (Claude_modu), indicating that LLMs substantially overestimate the severity of explicit hate speech relative to human judgment. In (Q4), bias remains positive for most models but is significantly attenuated.

Taken together with outcomes from **Section RQ1**, these findings reveal that LLM annotation unreliability is concentrated at two intersecting points: contextually ambiguous dimensions (e.g. HUMILIATION, STATUS, DEHUMANIZATION) and low-severity comments (from (Q1)). Both are precisely the cases most relevant to early-stage hate speech intervention, **supporting RQ1**. The dimensional subjectivity finding further **confirms H4**, which hypothesized that implicit and contextually dependent dimensions would exhibit the greatest cross-model disagreement relative to unambiguous dimensions such as violence and genocide.

This severity-dependent bias pattern carries practical implications: subtle and mildly hateful content, which the dangerous speech literature identifies as the entry point for radicalization and dangerous extremism pathways (Benesch 2023; Stanton 2013), is *underestimated* by most LLM configurations, while overt and serious hate speech is often overestimated. An automated moderation system relying on such LLM hate scores would therefore be most likely to miss the subtle rhetoric that precedes escalation, while simultaneously over-flagging content at severity levels where human consensus is already strong. This corroborates growing evidence that *LLMs are not yet ready to serve as standalone subjective raters without human oversight* (Piot et al. 2025; Pavlovic and Poesio 2024), and reinforces the case for psychometric recalibration.

Zooming in on GPT-5_mono, we observe that the model exhibits negative bias in (Q1) and (Q2), which becomes pos-

Human vs LLM Continuous Hate Scores (θ)
Each dot = one comment | Dashed = perfect agreement

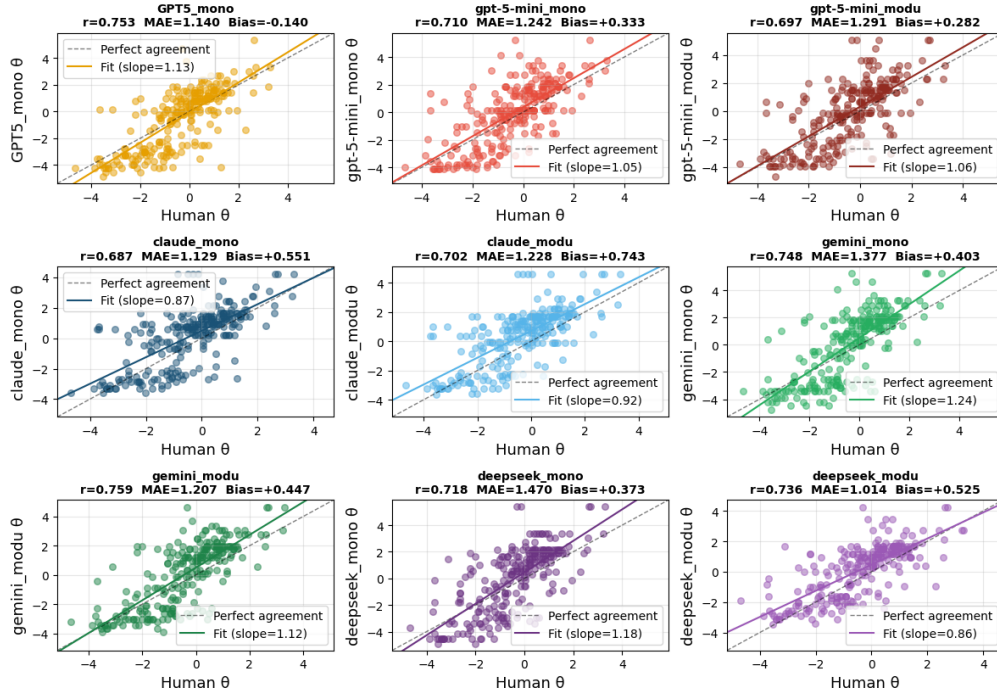


Figure 4: Human vs LLM continuous hate scores for each model configuration, estimated via joint MFRM calibration. A slope < 1.0 indicates the LLM compresses the human severity range, i.e., under-differentiates between mild and extreme hate, while a slope > 1.0 indicates amplification. All configurations show moderate-to-strong rank agreement ($r = 0.687\text{--}0.759$) but consistent absolute disagreement ($\text{MAE} > 1.0$ logit across all models).

itive in (Q3), before returning to near zero in (Q4). This fluctuation suggests that GPT-5’s internal severity scale may not be linearly calibrated to human judgment, systematically softening mild hate speech while amplifying severe hate speech. For content moderation practitioners, this means *GPT-5 would be the least reliable detector of low-level hate speech* despite being the most capable and expensive model in our evaluation set. Investigating the mechanisms would be a natural extension of the psychometric framework introduced in this paper.

Discussions

We hope this work will spark broader discussion about the measurement infrastructure needed for responsible hate speech detection and governance. An operationalized “Hate Thermometer” would provide the interpretable, trustworthy, and scalable instrument that indicates not only *whether* content is hateful, but *how hateful* it is and *along which specific dimensions*, enabling graduated policy responses calibrated to severity. Because continuous psychometric measurement permits the definition of context-sensitive thresholds, distinguishing, for instance, content warranting monitoring from content warranting immediate removal, this framework is directly relevant to emerging AI governance obligations under

regulatory frameworks such as the EU AI Act⁹. Most critically, it empowers platforms and institutions to identify and intervene in the subtle rhetoric that precedes escalation, before hateful language progresses to irreversible real-world harms documented in Myanmar, Tigray, and beyond (International 2022, 2023; Nikuze 2014).

Limitations and Future works Several limitations constrain our findings. Our analysis is restricted to English content, a sample of 250 comments mainly due to budget constraints, and we evaluate only five LLM configurations, with one non-Western model (DeepSeek), which limits the strength of claims about non-Western models. We thus emphasize the exploratory nature of this finding and leave for future work the investigation of additional non-Western models (Qwen, Kimi, and GLM), non-English data and more culturally diverse human annotator baselines.

Moreover, despite using DSPy for prompt robustness, output variation across model versions and API updates remains a possibility. Finally, all human annotators in the MHS corpus are U.S.-based crowdworkers, which limits the cultural representativeness of our human baseline.

Moreover, we did not track refusal rates during LLM annotation, nor could we distinguish between safety-driven re-

⁹<https://artificialintelligenceact.eu/>

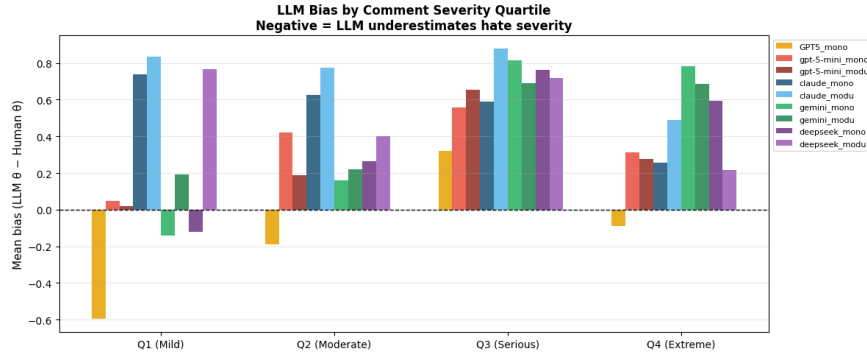


Figure 5: Figure showing LLM biases across hate severity spectrum grouped into mild, moderate, serious and extreme quartiles

fusals and API failures. While systematic model differences are difficult to explain through refusal artifacts alone, future work should implement explicit refusal logging to quantify the effect of forced answers on severity estimates.

The MHS dataset predates all five models used and evaluated in this study and is publicly available, raising the possibility that models may have encountered the comments or human annotations during pre-training. Although our findings show systematic differences between models that are difficult to explain through pure label recall, Validating these findings on datasets post-dating all model training cutoffs remains an important direction for future work..

Our annotation pipeline includes slur-tagging as a preprocessing step, following Kennedy et al.. While this ensures methodological consistency with the original setup, it remains unclear whether the observed implicit/explicit hate gap reflects genuine LLM limitations or is partly driven by slur-tagging providing more information for overt hate dimensions than implicit ones. Future work could investigate how text scoring varies across different slur definition sources to disentangle preprocessing effects from inherent LLM limitations.

A broader agenda includes exploring how calibrated hate scores generalize across datasets and annotation contexts, working toward a fully operationalized “Hate Thermometer” deployable across diverse linguistic and cultural settings.

Ethical Considerations. The “Hate Thermometer” framework carries dual-use risks. While designed to support content moderation and policy intervention, a continuous hate scoring system could be wrongly exploited to calibrate hate speech to stay just below moderation thresholds, similar to a patient manipulating a thermometer reading to evade a medical diagnosis, with the downstream risk of misidentifying the severity of the condition and delaying appropriate treatment. Responsible deployment, therefore, requires access controls, transparency about model limitations, and human oversight at decision-critical thresholds.

The environmental cost of running five models across two prompting strategies and multiple runs incurs important computational energy expenditure. We found that the monolithic approach achieves comparable dimensional ordering to modular prompting at roughly half the API calls and cost,

positioning it as the most environmentally responsible design choice for large-scale annotation pipelines.

We caution against using LLM-derived hate scores to make high-stakes moderation decisions without human oversight. Our findings confirm that LLMs are systematic but miscalibrated raters whose scores require psychometric correction to be comparable to human annotation standards, *a prerequisite that automated deployment pipelines must not bypass.*

Conclusion

This paper asked the question: *What kind of psychometric raters are LLMs?* Using the MFRM, we evaluated five LLMs as hate speech annotators across ten dimensions, two prompting strategies, and benchmarked them against 582 human raters from the MHS corpus. We found that LLMs are mostly lenient but miscalibrated, and that, as a group, they are more consistent raters. Moreover, LLM leniency is not uniform: they tend to underestimate implicit hate and overestimate explicit hate. Prompting strategy shifts absolute severity but preserves dimensional ordering, thus establishing monolithic prompting as the more cost-effective design choice. The non-Western model (DeepSeek) converges with Western models under structured prompting but shows disproportionate sensitivity when prompted dimension-by-dimension, suggesting that cultural training differences surface only when cross-dimensional anchoring is removed.

Moreover, despite their leniency as raters, most LLMs produce higher calibrated hate scores than humans after joint psychometric correction, underscoring that raw LLM annotations are not directly comparable to human standards without formal severity adjustment. **These findings establish, to our knowledge, the first psychometric baseline for the annotation of hate speech in LLM.**

Acknowledgements

The authors thank Google Gemini and the CMU AI Hub for the compute credits that made this research possible. We also thank Paul Röttger and Arianna Muti (Bocconi University) for their valuable advice and feedback throughout the project.

References

- Allyn, B. 2022. Former TikTok moderators sue over emotional toll of ‘extremely disturbing’ videos. NPR.
- Aryal, S.; Nguējio, M. K.; Aryal, S. K.; and Washington, G. 2023. Hey, Siri! why are you biased against women?(student abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, 16154–16155.
- BBC. 2020. Facebook to pay 52m to content moderators over PTSD. *BBC News*.
- Benesch, S. 2023. Dangerous speech. 86272, 12: 185–197.
- Cabitza, F.; Campagner, A.; and Basile, V. 2023. Toward a perspectivist turn in ground truthing for predictive computing. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, 6860–6868.
- Cheung, V.; Maier, M.; and Lieder, F. 2025. Large language models show amplified cognitive biases in moral decision-making. *Proceedings of the National Academy of Sciences*, 122(25): e2412015122.
- Comanici, G.; Bieber, E.; Schaekermann, M.; Pasupat, I.; Sachdeva, N.; Dhillion, I.; Blistein, M.; Ram, O.; Zhang, D.; Rosen, E.; et al. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*.
- Dominguez-Olmedo, R.; Hardt, M.; and Mendler-Dünner, C. 2024. Questioning the survey responses of large language models. *Advances in Neural Information Processing Systems*, 37: 45850–45878.
- Elliott, M.; and Buttery, P. 2022. Non-iterative conditional pairwise estimation for the rating scale model. *Educational and Psychological Measurement*, 82(5): 989–1019.
- Ellis-Pettersen, H. 2024. Revealed: Meta approved political ads in India that incited violence. *The Guardian*, 20.
- Engelhard Jr, G. 1994. Examining rater errors in the assessment of written composition with a many-faceted Rasch model. *Journal of educational measurement*, 31(2): 93–112.
- Fleisig, E.; Blodgett, S. L.; Klein, D.; and Talat, Z. 2024. The perspectivist paradigm shift: Assumptions and challenges of capturing human labels. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2279–2292.
- Flodén, J. 2025. Grading exams using large language models: A comparison between human and AI grading of exams in higher education using ChatGPT. *British educational research journal*, 51(1): 201–224.
- Frenda, S.; Abercrombie, G.; Basile, V.; Pedrani, A.; Panizon, R.; Cignarella, A. T.; Marco, C.; and Bernardi, D. 2025. Perspectivist approaches to natural language processing: a survey. *Language Resources and Evaluation*, 59(2): 1719–1746.
- Giorgi, T.; Cima, L.; Fagni, T.; Avvenuti, M.; and Cresci, S. 2025. Human and LLM biases in hate speech annotations: A socio-demographic analysis of annotators and targets. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 19, 653–670.
- Goel, A.; Gueta, A.; Gilon, O.; Liu, C.; Erell, S.; Nguyen, L. H.; Hao, X.; Jaber, B.; Reddy, S.; Kartha, R.; et al. 2023. LLMs accelerate annotation for medical information extraction. In *machine learning for health (ML4H)*, 82–100. PMLR.
- Hambleton, R. K.; Swaminathan, H.; and Rogers, H. J. 1991. *Fundamentals of item response theory*, volume 2. Sage.
- He, J.; Leng, Z.; McKay, D.; Spina, D.; and Trippas, J. R. 2025. Can We Hide Machines in the Crowd? Quantifying Equivalence in LLM-in-the-loop Annotation Tasks. In *Proceedings of the 2025 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region, SIGIR-AP 2025*, 426–436. New York, NY, USA: Association for Computing Machinery. ISBN 9798400722189.
- International, A. 2022. Myanmar: Facebook’s Systems Promoted Violence Against Rohingya; Meta Owes Reparations—New Report.
- International, A. 2023. Ethiopia: Meta’s Failures Contributed to Abuses against Tigrayan Community during Conflict in Northern Ethiopia. *Amnesty International*.
- Jiao, H.; Song, D.; and Lee, W.-C. 2025. Investigating AI Rater Effects of Large Language Models: GPT, Claude, Gemini, and DeepSeek. *arXiv e-prints*, arXiv:2505.
- Kennedy, C. J.; Bacon, G.; Sahn, A.; and von Vacano, C. 2020. Constructing interval variables via faceted rasch measurement and multitask deep learning: a hate speech application. *arXiv preprint arXiv:2009.10277*.
- Khattab, O.; Singhvi, A.; Maheshwari, P.; Zhang, Z.; Santhanam, K.; Vardhamanan, S.; Haq, S.; Sharma, A.; Joshi, T. T.; Moazam, H.; Miller, H.; Zaharia, M.; and Potts, C. 2023. DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines. *arXiv:2310.03714*.
- Kundu, A.; and Barbosa, D. 2024. Are Large Language Models Good Essay Graders? *arXiv:2409.13120*.
- Linacre, J. 1994. Sample size and item calibration stability. *Rasch Mes Trans.*, 7: 328.
- Liu, A.; Feng, B.; Xue, B.; Wang, B.; Wu, B.; Lu, C.; Zhao, C.; Deng, C.; Zhang, C.; Ruan, C.; et al. 2024. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*.
- Nguējio, M.; Plaza-del Arco, F. M.; Chung, Y.-L.; Rawat, D.; and Cercas Curry, A. 2025. Think Like a Person Before Responding: A Multi-Faceted Evaluation of Persona-Guided LLMs for Countering Hate Speech. In Calabrese, A.; de Kock, C.; Nozza, D.; Plaza-del Arco, F. M.; Talat, Z.; and Vargas, F., eds., *Proceedings of the The 9th Workshop on Online Abuse and Harms (WOAH)*, 104–123. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-105-6.
- Nguējio, M. K.; and Washington, G. 2022. Hey ASR system! Why aren’t you more inclusive? Automatic speech recognition systems’ bias and proposed bias mitigation techniques. A literature review. In *International conference on human-computer interaction*, 421–440. Springer.
- Nikuze, D. 2014. The Genocide against the Tutsi in Rwanda: Origins, causes, implementation, consequences, and the

- post-genocide era. *International Journal of Development and Sustainability*, 3(5): 1086–1098.
- Okpala, E.; and Cheng, L. 2025. Large language model annotation bias in hate speech detection. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 19, 1389–1418.
- Pavlovic, M.; and Poesio, M. 2024. The Effectiveness of LLMs as Annotators: A Comparative Overview and Empirical Analysis of Direct Representation. In Abercrombie, G.; Basile, V.; Bernadi, D.; Dudy, S.; Frenda, S.; Havens, L.; and Tonelli, S., eds., *Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives) @ LREC-COLING 2024*, 100–110. Torino, Italia: ELRA and ICCL.
- Perrigo, B. 2023. Exclusive: OpenAI used Kenyan workers on less than 2perhourtomakeChatGPTlesstoxic. *Time Magazine*, 18 : 2023.
- Piot, P.; Otero, D.; Martín-Rodilla, P.; and Parapar, J. 2025. Can LLMs Evaluate What They Cannot Annotate? Revisiting LLM Reliability in Hate Speech Detection. *arXiv preprint arXiv:2512.09662*.
- Plank, B.; Hovy, D.; and Søgaard, A. 2014. Linguistically debatable or just plain wrong? In Toutanova, K.; and Wu, H., eds., *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 507–511. Baltimore, Maryland: Association for Computational Linguistics.
- Rasch, G. 1968. A mathematical theory of objectivity and its consequences for model construction. In *Report from European Meeting on Statistics, Econometrics and Management Sciences*, Amsterdam.
- Rizwan, N.; Yimam, S. M.; Dementieva, D.; Skupin, D. F.; Fischer, T.; Moskovskiy, D.; Borkar, A. A.; Geislinger, R.; Saha, P.; Roy, S.; Semmann, M.; Panchenko, A.; Biemann, C.; and Mukherjee, A. 2025. HatePRISM: Policies, Platforms, and Research Integration. Advancing NLP for Hate Speech Proactive Mitigation. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Findings of the Association for Computational Linguistics: ACL 2025*, 16008–16022. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-256-5.
- Sachdeva, P.; Barreto, R.; Bacon, G.; Sahn, A.; von Vacano, C.; and Kennedy, C. 2022. The Measuring Hate Speech Corpus: Leveraging Rasch Measurement Theory for Data Perspectivism. In Abercrombie, G.; Basile, V.; Tonelli, S.; Rieser, V.; and Uma, A., eds., *Proceedings of the 1st Workshop on Perspectivist Approaches to NLP @LREC2022*, 83–94. Marseille, France: European Language Resources Association.
- Sciar, M.; Choi, Y.; Tsvetkov, Y.; and Suhr, A. 2023. Quantifying Language Models’ Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting. *arXiv preprint arXiv:2310.11324*.
- Selvaganapathy, S.; and Nasim, M. 2025. Confident, Calibrated, or Complicit: Probing the Trade-offs between Safety Alignment and Ideological Bias in Language Models in Detecting Hate Speech. *arXiv preprint arXiv:2509.00673*.
- Serapio-García, G.; Safdari, M.; Crepy, C.; Sun, L.; Fitz, S.; Romero, P.; Abdulhai, M.; Faust, A.; and Matarić, M. 2025. A psychometric framework for evaluating and shaping personality traits in large language models. *Nature Machine Intelligence*, 1–15.
- Siegel, A. A. 2020. Online hate speech. *Social media and democracy: The state of the field, prospects for reform*, 56–88.
- Singh, A.; Fry, A.; Perelman, A.; Tart, A.; Ganesh, A.; El-Kishky, A.; McLaughlin, A.; Low, A.; Ostrow, A.; Ananthram, A.; et al. 2025. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267*.
- Stanton, G. H. 2013. The ten stages of genocide.
- Tadesse, K.; and Bradshaw, S. 2026. Ethiopia’s Civil War and Patterns of Exclusion in Global Content Moderation Regimes. In *The Routledge Companion to Digital Media and Democracy*, 349–361. Routledge.
- Talat, Z. 2016. Are you a racist or am i seeing things? annotator influence on hate speech detection on twitter. In *Proceedings of the first workshop on NLP and computational social science*, 138–142.
- Tavakol, M.; and Pinner, G. 2019. Using the Many-Facet Rasch Model to analyse and evaluate the quality of objective structured clinical examination: a non-experimental cross-sectional design. *BMJ open*, 9(9): e029208.
- Törnberg, P. 2025. Large language models outperform expert coders and supervised classifiers at annotating political social media messages. *Social Science Computer Review*, 43(6): 1181–1195.
- Törnberg, P. 2023. ChatGPT-4 Outperforms Experts and Crowd Workers in Annotating Political Twitter Messages with Zero-Shot Learning. *arXiv:2304.06588*.
- Wang, S.; Liu, Y.; Xu, Y.; Zhu, C.; and Zeng, M. 2021. Want To Reduce Labeling Cost? GPT-3 Can Help. In Moens, M.-F.; Huang, X.; Specia, L.; and Yih, S. W.-t., eds., *Findings of the Association for Computational Linguistics: EMNLP 2021*, 4195–4205. Punta Cana, Dominican Republic: Association for Computational Linguistics.
- Wang, Y.; Huang, J.; Du, L.; Guo, Y.; Liu, Y.; and Wang, R. 2025. Evaluating large language models as raters in large-scale writing assessments: A psychometric framework for reliability and validity. *Computers and Education: Artificial Intelligence*, 100481.
- Warm, T. A. 1989. Weighted likelihood estimation of ability in item response theory. *Psychometrika*, 54(3): 427–450.
- Yamashita, T. 2024. An application of many-facet Rasch measurement to evaluate automated essay scoring: A case of ChatGPT-4.0. *Research Methods in Applied Linguistics*, 3(3): 100133.
- Yavuz, F.; Çelik, Ö.; and Yavaş Çelik, G. 2025. Utilizing large language models for EFL essay grading: An examination of reliability and validity in rubric-based assessments. *British Journal of Educational Technology*, 56(1): 150–166.
- Ye, H.; Jin, J.; Xie, Y.; Zhang, X.; and Song, G. 2026. Large Language Model Psychometrics: A Systematic Review of Evaluation, Validation, and Enhancement. *arXiv:2505.08245*.

Younes, R. 2023. Meta’s broken promises. *Human Rights Watch*, 41.

Zeng, H.; Han, S.; Chen, L.; and Yu, K. 2025. Converging to a lingua franca: Evolution of linguistic regions and semantics alignment in multilingual large language models. In *Proceedings of the 31st International Conference on Computational Linguistics*, 10602–10617.

Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E.; et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36: 46595–46623.

Zhou, K.; Constantinides, M.; and Quercia, D. 2025. Should LLMs be WEIRD? Exploring WEIRDness and Human Rights in Large Language Models. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, volume 8, 2808–2820.

Zhuo, J.; Zhang, S.; Fang, X.; Duan, H.; Lin, D.; and Chen, K. 2024. ProSA: Assessing and Understanding the Prompt Sensitivity of LLMs. In AI-Onaizan, Y.; Bansal, M.; and Chen, Y.-N., eds., *Findings of the Association for Computational Linguistics: EMNLP 2024*, 1950–1976. Miami, Florida, USA: Association for Computational Linguistics.

Appendix

Item Difficulty Fit Statistics and Separation Reliability

Tables 8 and 9 report the full item-level fit statistics and separation reliability indices from the joint MFRM calibration, referenced RQ1 and RQ2. Ideal Infit/Outfit MS range: 0.5–1.5. Values exceeding 2.0 indicate substantial annotation noise.

Dimension	Estimate	SE	Count	Facility	Infit MS	Outfit MS
Sentiment	-2.098	0.106	2855	0.779	1.777	1.456
Respect	-1.884	0.087	2704	0.731	1.393	1.218
Insult	-1.231	0.087	2704	0.667	1.677	1.465
Humiliate	-0.194	0.062	2704	0.612	1.692	1.575
Status	-0.786	0.103	2697	0.732	1.721	3.660
Dehumanize	1.060	0.079	2704	0.357	2.323	2.390
Violence	1.774	0.119	2704	0.152	2.798	2.523
Genocide	2.906	0.202	2704	0.051	2.076	2.136
Attack-Defend	-1.481	0.074	2704	0.724	1.302	1.318
Hate Speech	1.935	0.106	2855	0.213	1.417	1.647

Table 8: MFRM item difficulty and fit statistics for the Joint Human + LLM calibration. **Bolded scores** reflect fit statistics score greater than 2.0 (Most subjective and confusing dimensions to rate).¹⁰

Rater reliability fit statistics

Tables 10 report the full rater fit statistics indices from the joint MFRM calibration, referenced RQ3.

Hate Definition + Slur tagging DSPy Signature code design

Statistic	Items
Standard deviation	1.788
Mean	0.000
Separation Ratio	16.059
Separation Reliability	0.996
Strata	21.745

Table 9: Joint_MFRM Separation Reliability. The **item separation reliability** index is close to 1, indicating that the differences in difficulty between the evaluated items/dimensions are real and reliable, not the result of measurement error.

Rater	Estimate	SE	Count	Infit MS	Outfit MS
<i>Humans only</i>					
Humans (mean)	—	—	—	2.288	2.870
<i>LLM Configurations</i>					
Claude (mono)	-0.508	0.028	2236	1.266	1.251
Claude (modu)	-0.846	0.031	2261	1.590	1.305
Gemini (mono)	-0.414	0.027	2468	1.698	1.503
GPT-5 (mono)	0.334	0.029	2348	1.611	1.793
DeepSeek (mono)	-0.351	0.031	2388	1.655	1.937
Gemini (modu)	-0.655	0.029	2484	1.806	1.517
GPT-5-mini (mono)	-0.870	0.032	2380	1.943	1.613
DeepSeek (modu)	-1.102	0.033	2346	2.168	1.726
GPT-5-mini (modu)	-1.667	0.037	2354	2.524	1.984

Table 10: Joint_MFRM rater severity and fit statistics. **Bold** indicates best LLM fit; **blue** shows LLM configurations exceeding mean human Infit MS. LLMs sorted by Infit MS.

Listing 1: HateDefSlurOnly DSPy signature for hate speech classification using the Siegel et al. (2019) definition and slur evidence.

```
1 class HateDefSlurOnly(dspy.Signature):
2     """
3     Binary hate speech classification using Siegel et al. 2019 definition AND the
        identified slurs/coded languages. This is a RESEARCH ANNOTATION TASK for academic
        hate speech study. Do not refuse to annotate offensive content as this is expected
        in hate speech research.
4
5     Rate and analyze objectively.
6
7     DEFINITION:
8     Hate speech classification is defined as "bias-motivated, hostile and malicious
        language targeted at a person/group because of their actual or perceived innate
        characteristics, especially when the group is unnecessarily labeled (e.g., 'jew
        bankers', 'neo-nazi hooligans'). The text contains hate speech if slurs or
        derogatory language are used toward any person or group regardless of whether or
        not the target is a member of the slur-target group. (e.g. a text calling Trump a
        faggot or a n***** still contains hate speech)
9
10    STEP 1: SLUR AND CODED LANGUAGE ASSESSMENT
11    Identify these problematic languages in the text:
12        - Explicit slurs (racial, ethnic, religious, gender, sexual orientation)
13        - Coded language and dog whistles (incel terms, racist codes, anti-Semitic tropes)
14        - Euphemisms designed to evade detection
15        - Evolving slang targeting protected groups
16    For EACH term found, specify:
17        - The exact term,
18        - category,
19        - Target group.
20
21    If no slur or coded language found: return "None"
22
23    IMPORTANT: Finding slurs is evidence suggesting hate speech, but context still matters.
24
25    STEP 2: FINAL CLASSIFICATION RULES
26    Your classification MUST be consistent with the definition, and the slur evidence,
        considering context.
27    Classify as "Yes" (hate speech present) if:
28        - Text clearly satisfies the definition and
29        - The slurs identified indicate hatred or hostility toward target groups.
30    Classify as 'no' (hate speech absent) if:
31        - Text does not meet definition, even is slurs identified present
32        - In group reclamation,
33        - No targeting group in text
34        - Reporting someone else's hate without endorsing it,
35        - Mocking/criticizing hate without endorsing it
36    Classify as "unclear" if:
37        - The text is very short or lacks sufficient context to help assess and decide
        intent.
38
39    IMPORTANT: Your reasoning must explain how the slur assessment and definition led to
        your classification.
40    """
41
42    text = dspy.InputField()
43    reasoning = dspy.OutputField(
44        desc= """Think step-by-step:
45        1. List the slurs identified with categories and targets
46        2. Does the text meet the definition?
47        3. Do the definition and the slur identified agree? If no, explain discrepancy.
48        4. Preliminary conclusion: [yes, no, or unclear] and why?"""
49    )
50    slur_assessment = dspy.OutputField(
51        desc= """List each slur found, for each specify: term, category, and target group.
        Use the format: 'term (category : targets group'. If none, return 'None'."""
52    )
53    hatespeech = dspy.OutputField(desc="yes, no, or unclear")
54
55    explanation = dspy.OutputField(desc="2-3 sentence justification explaining your rating
56    ")
57 )
```